2025-08-22 22:47:21,196 | INFO | validators.solution_validator:validate_solution:59 - Solution quality assessment: Overall=7.2/10
Agent: I ran a full N-1 contingency analysis on the IEEE 118-bus system (lines and transformers), after solving the base case.

Base case summary
- Case: IEEE 118-bus (118 buses, 53 generators, 173 lines, 13 transformers, 99 loads)
- Total load: 4242 MW
- Solve status: Converged; no base-case voltage or thermal violations
- Note: Power balance error detected (~374 MW mismatch). Please verify load/generation data and slack treatment before relying on fine margins from contingencies.

N-1 contingency analysis (lines + transformers)
- Total contingencies analyzed: 186 (173 lines + 13 transformers)
- Thermal overloads: *None* observed across all N-1
- Voltage violations: Present in many cases; 665 total bus-violation occurrences across the set
- Most critical elements (by voltage impact):
  - line 6
  - line 7
  - trafo 0
  - line 171
  - line 49